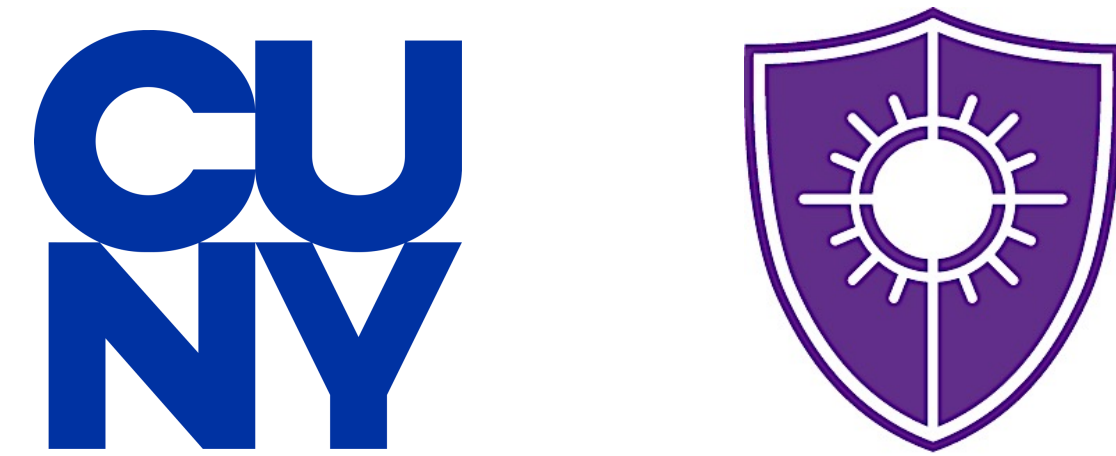




Higher Quality Narrative Results in Better Video Gaming Experience but Only for Higher Quality Graphics.

Michael Vail¹, Claire Corwin², Aidan Morris², Jane Connelly², & Anurag Rimzhim³
¹CUNY Graduate Center, ²College of the Holy Cross, ³Stevens Institute of Technology



INTRODUCTION

Literature Review

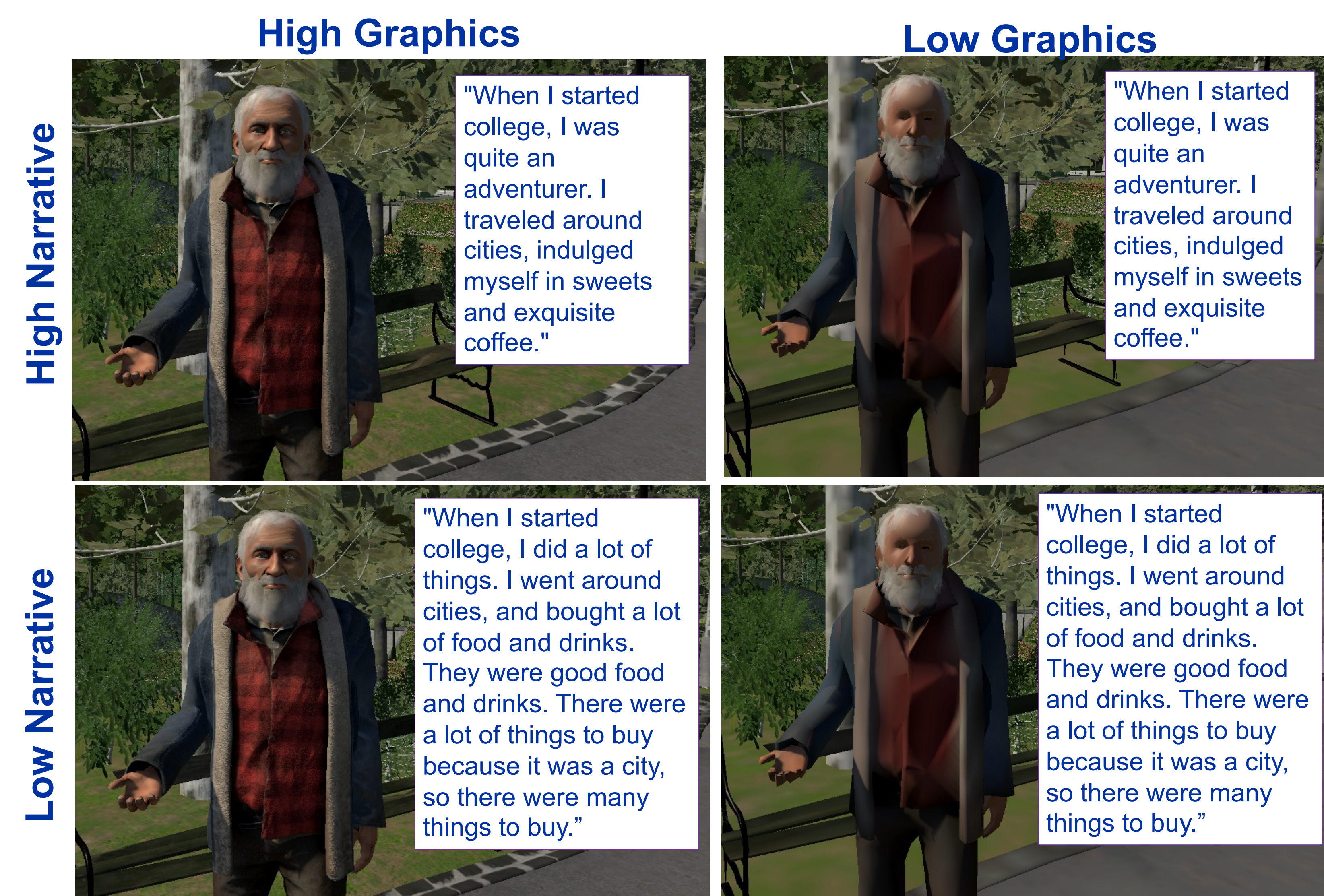
- The use of virtual reality has expanded rapidly, with creators increasingly relying on narrative-driven environments to shape user experience through transportation and interactivity. Two key elements that influence VR engagement are **narrative quality** and **graphics quality**.
- Narrative-based media aim to enhance user engagement by increasing **presence** and **enjoyment** (Chang, 2025; Kim et al., 2021).
- Prior research shows an interplay between narrative and graphics: *visually appealing graphics can capture attention even when narrative content is weak* (Newman et al., 2022), while *strong narratives can sustain engagement even with minimal visual detail* (Heider & Simmel, 1944).

Research Hypotheses

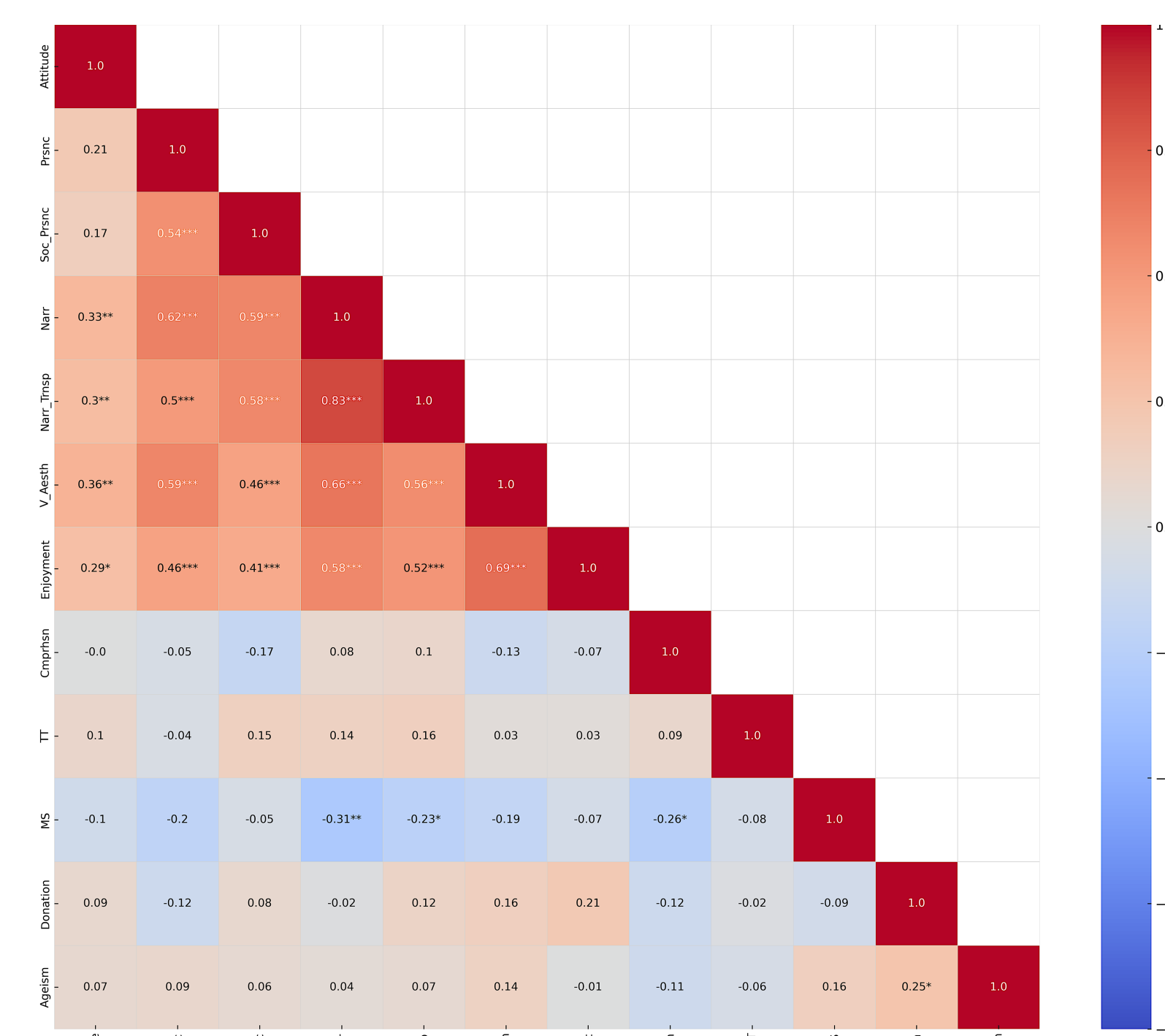
- High **narrative quality** will produce a better **user experience** than low narrative quality.
- High **graphics quality** will produce a better **user experience** than low graphics quality.

METHOD

- 75 undergraduates** (M = 20.29 yrs; SD = 2.35) played the **VR game** on a desktop with **1080p 32" monitor**
- Design:** 2 × 2 (Narrative [High vs. Low] X Graphics [High vs. Low])
- Measures:** Experiential, Persuasion-Based, Behavioral, Motion Sickness

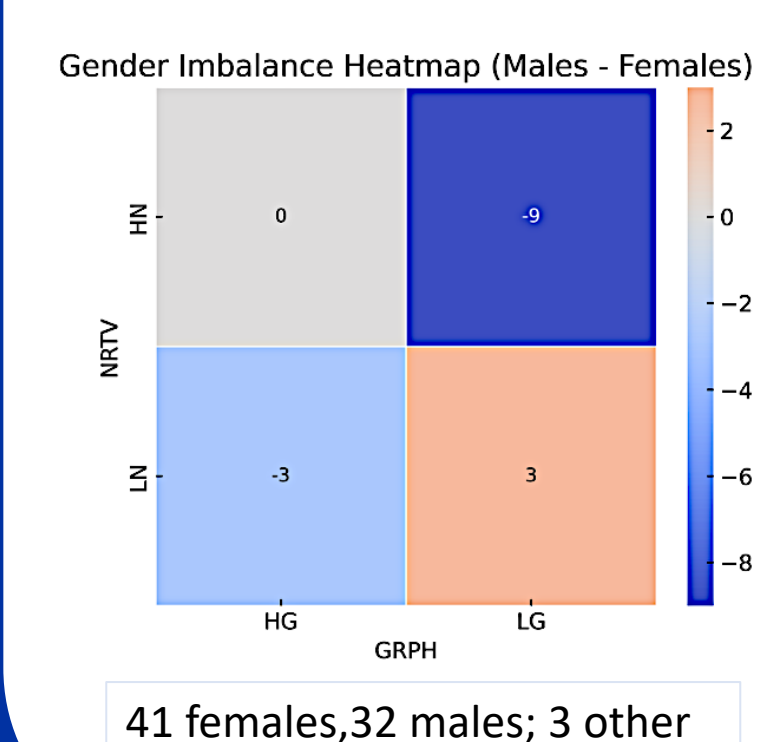


RESULTS

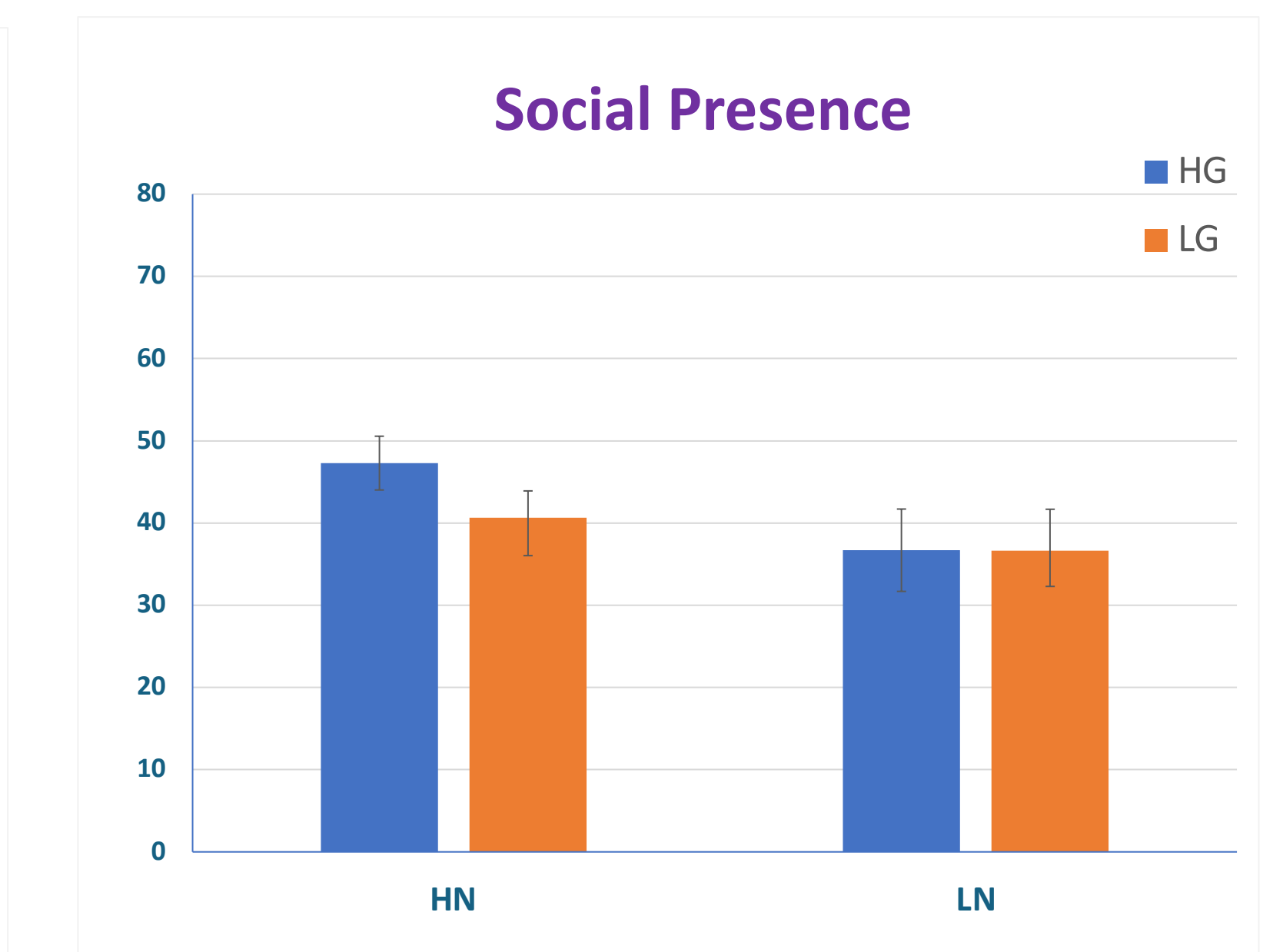
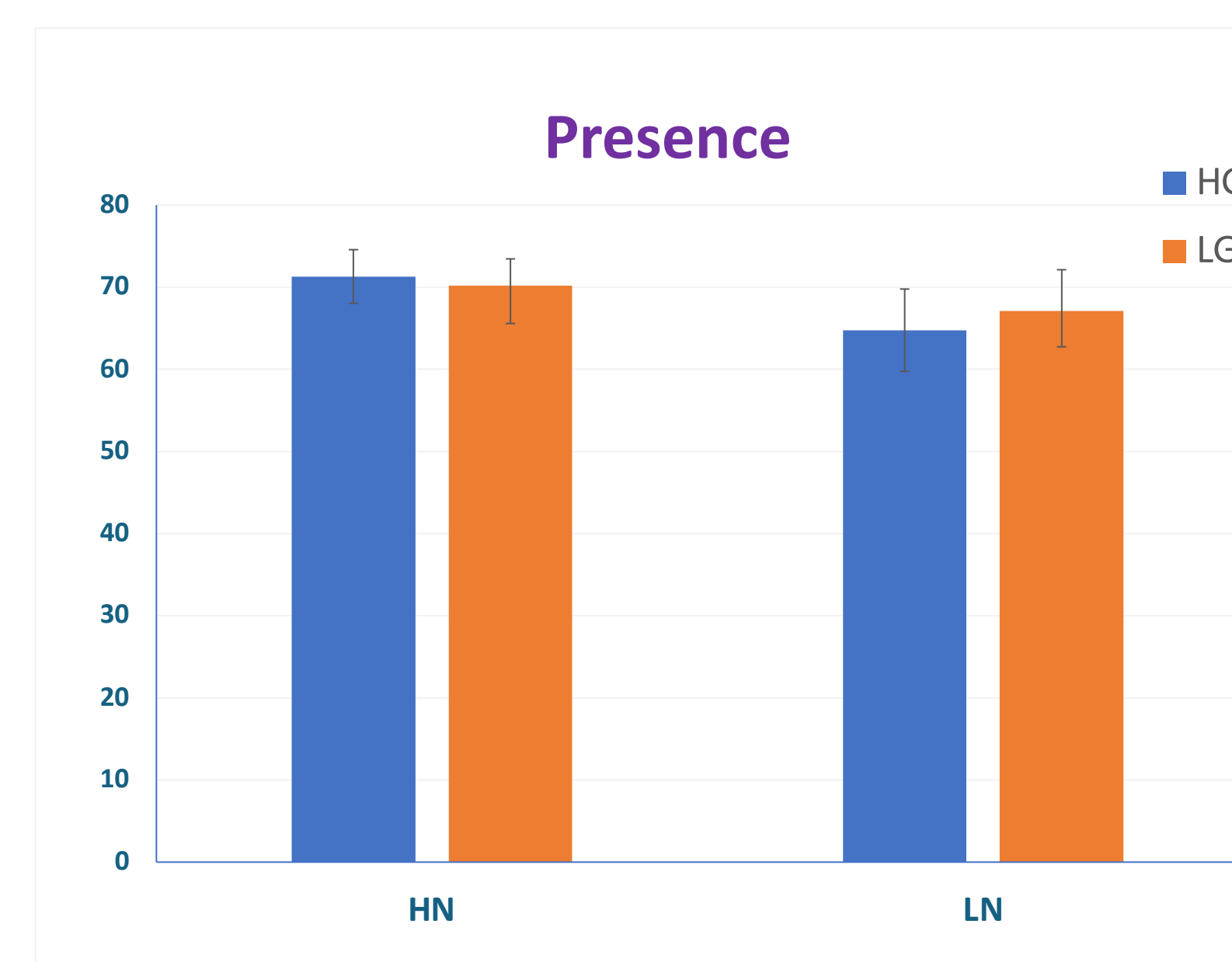
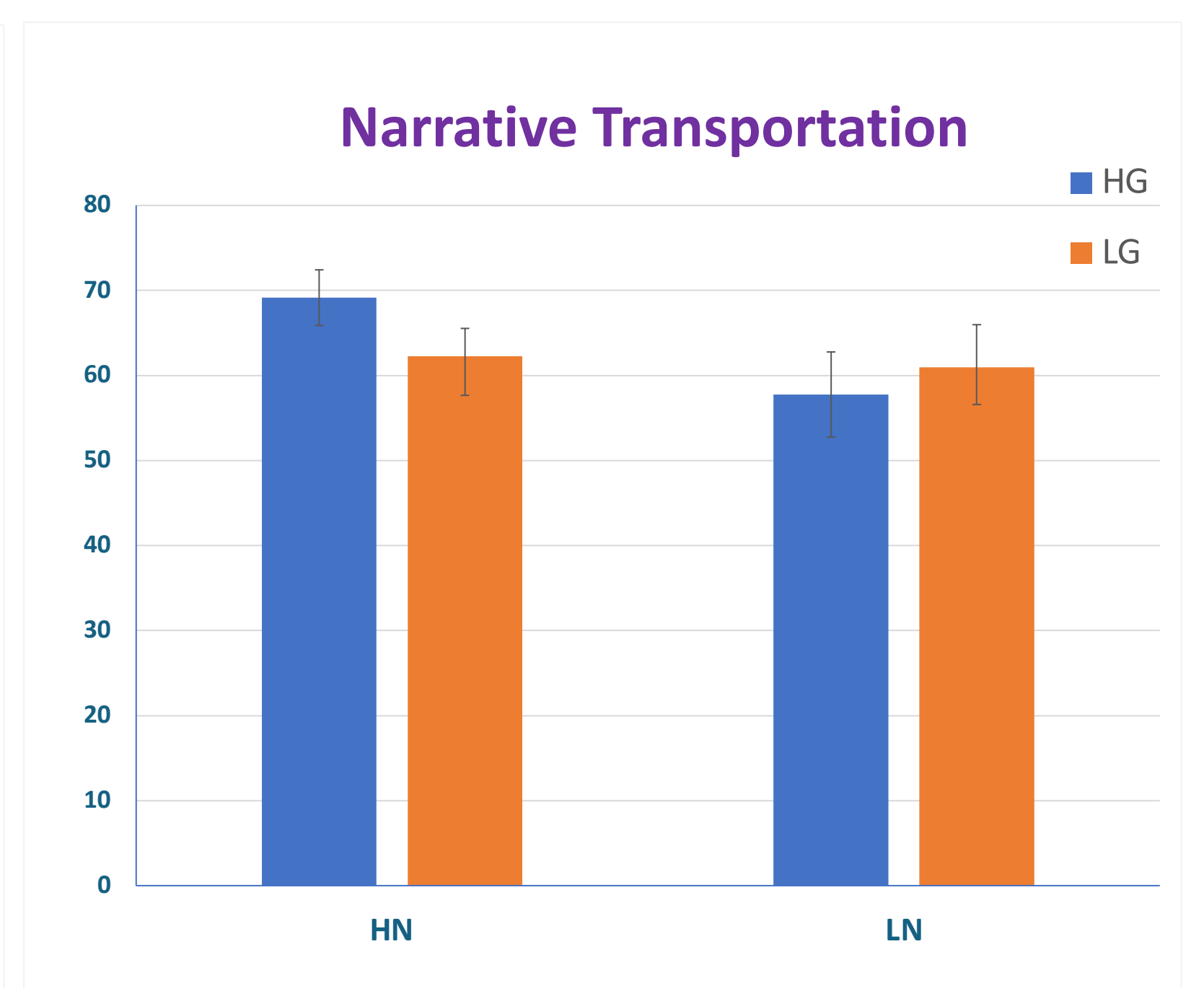
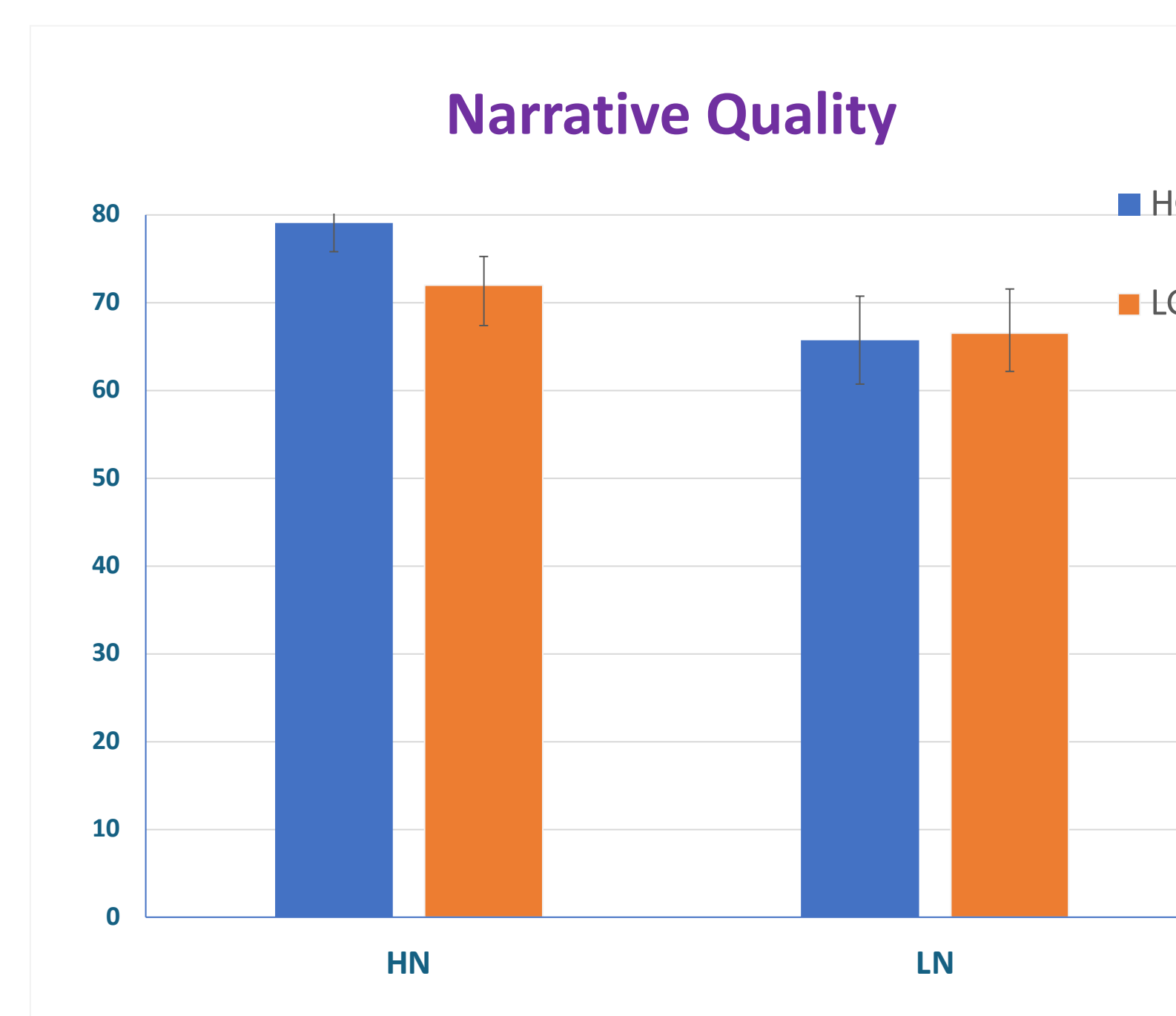


Clustered heatmap of **correlations** among **experiential** (narrative engagement, transportation, presence, social presence, aesthetics, enjoyment), **persuasion-related** (attitudes, donation intentions, ageism, comprehension), and **behavioral variables** (time taken), and the covariate **motion sickness** (MS).

- Experiential variables** were **moderately to strongly correlated** ($r = .50 - .80$)
- Motion-sickness (MS)** was used as a covariate:
 - Consistent negative correlations with nearly all experiential variables ($r \approx -.30$ to $-.45$),
 - No meaningful correlation with the two IVs.



- Compared to males, **females** reported significantly higher:
 - Attitude** ($d = -0.70$)
 - Visual Aesthetics** ($d = -0.70$)
 - Presence** ($d = -0.63$)
 - Enjoyment** ($d = -0.64$)
- Based on Welch-corrected independent sample t-test ($p < .05$)



- A multivariate analysis of covariance (**MANCOVA**) was conducted to examine the effects of **Graphics Quality** (high vs. low) and **Narrative Quality** (high vs. low) on **six experiential outcomes**, while **statistically controlling for Motion Sickness (MS)**.
 - Main effect of Narrative Quality:** Wilks's $\Lambda = .809$, $F(6, 64) = 2.52$, $p = .028$
 - Not significant:** Main effect of **Graphics Quality** or Interaction ($p > .05$)
 - MS had a strong multivariate effect**, Wilks's $\Lambda = .727$, $F(6, 64) = 4.01$, $p = .002$
- ANOVAs** were performed on **persuasion-related outcomes** and **behavioral variables**.
 - None of the non-experiential variables were affected by the IVs ($p > .05$).

DISCUSSION

- Narrative Quality** consistently **enhanced experiential outcomes** (involving narrative engagement, transportation, presence, and social presence) of playing a narrative-based **desktop VR game**, even when controlling for motion sickness.
 - These findings align with theories of narrative immersion and transportation, which argue that rich and coherent narratives foster deeper cognitive and emotional involvement (Green & Brock, 2000).
- Graphics Quality** did not significantly influence experiential or persuasion outcomes, suggesting that **in narrative-based media, story richness may carry more psychological weight than visual fidelity**.
 - This aligns with research indicating that **once a minimum visual threshold is met**, narrative coherence becomes the primary driver of engagement (Slater, 2003).
- Gender differences** may highlight baseline tendencies and possible mechanisms that shape **stronger experiential effects among females**.

Conclusion: **Narrative quality plays a dominant role in shaping immersive experiences** and underscore the importance of **individual differences** (e.g., motion sickness sensitivity) in determining how people respond to immersive media.

KEY TAKEAWAYS

- Narrative quality is a primary driver of VR experience.**
- Graphics quality matters only when narratives are strong.**
- Motion sickness** suppresses experiential outcomes.
- Females** report consistently higher presence, aesthetics, and enjoyment.
- Persuasion outcomes** were not influenced by narrative or graphics.